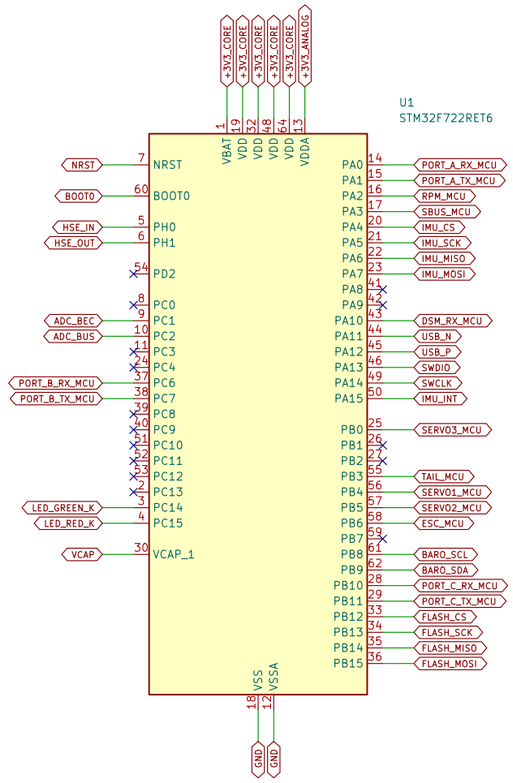
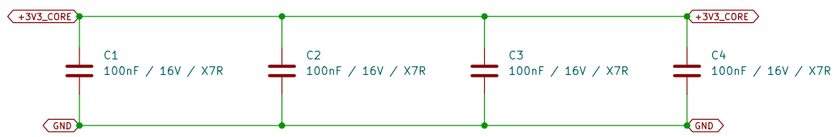
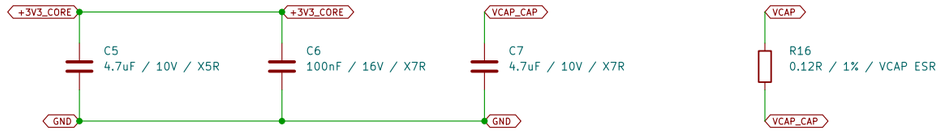



U1 stacked VSS pins: 18, 31, 47, 63

CORE BYPASS: place one 100 nF at each VDD pin

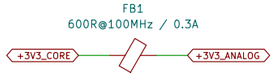


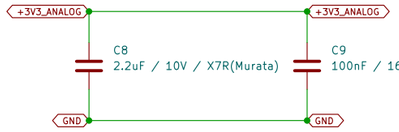







LQFP64 VCAP: 4.7 uF with total ESR 0.1–0.2 ohm
R16 + capacitor ESR + copper must meet this range
Do not connect VCAP to the 3.3 V supply







CLASSIC NEXUS TARGET COMPATIBILITY
UART A/B/C RX and TX follow stock target resources
PA9 is reserved but not exposed; PA10 = DSM RX
Unassigned MCU pins intentionally unconnected

BOOT0 button pulls to 3.3 V for ROM DFU
8 MHz HSE required by stock F7X2 startup
See linked sheets for complete subsystem circuits

Sensors + Blackbox
File: sensors.kicad_sch

USB + Clock + Service
File: usb–clock.kicad_sch

io–ports
File: io–ports.kicad_sch

voltage–sense
File: voltage–sense.kicad_sch

power–bec
File: power–bec.kicad_sch

power–core
File: power–core.kicad_sch

power–usb–dsm
File: power–usb–dsm.kicad_sch

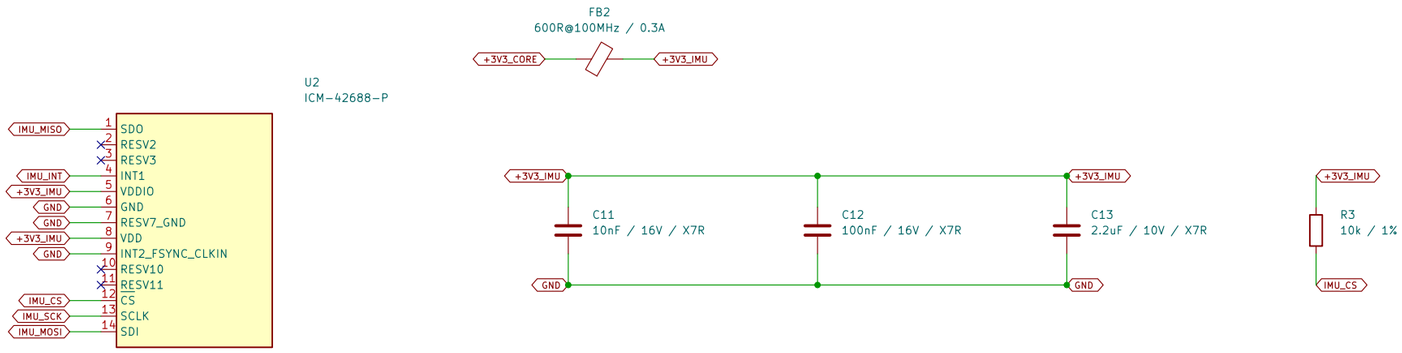
input–levels
File: input–levels.kicad_sch

ENGINEERING DEVELOPMENT – NOT FOR FABRICATION OR FLIGHT

Sheet: /
File: f722–heli.kicad_sch

**Title: F722 HELICOPTER CONTROLLER / MCU + VCAP**

| Size: A3 | Date: 2026–10–04 | **Rev: A–dev** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/9 |





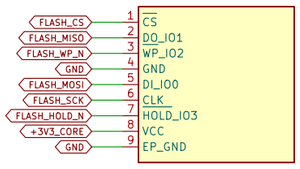


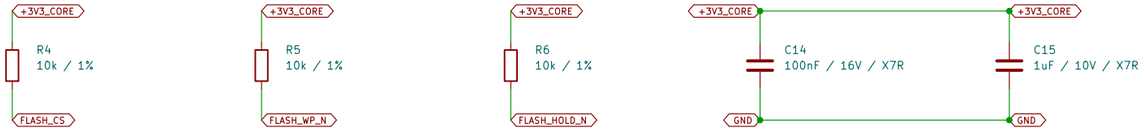




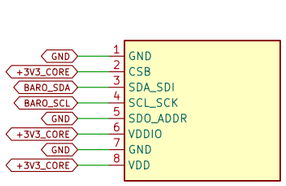


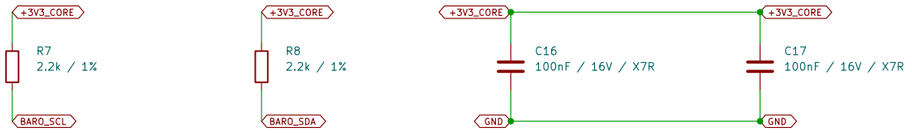
U2: C11 at pin 5; C12 + C13 at pin 8
CW90 stock transform: corrected X=raw Y, Y=−raw X, Z=raw Z
Physical sensor +Y must point toward board forward arrow
INT1 active−high push−pull; no external pull required




Flash exposed pad may float or connect GND per Winbond
GND chosen; no open vias in exposed pad (avoid solder wicking)



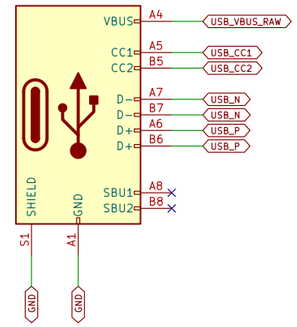

DPS368: user−approved replacement for unavailable SPL06
SDO low = 0x76; CSB high = I2C; clockwise pin numbering
Pressure vent remains open; stock RF driver bring−up pending

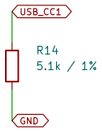
ENGINEERING DEVELOPMENT — NOT FOR FABRICATION OR FLIGHT

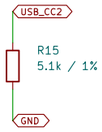
Sheet: /Sensors + Blackbox/
File: sensors.kicad_sch

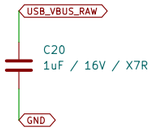
**Title: F722 HELICOPTER CONTROLLER / SENSORS + BLACKBOX**

| Size: A3 | Date: 2026−10−04 | **Rev: A−dev** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/9 |









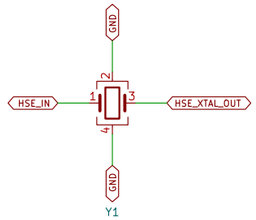


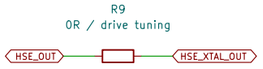





J1 VBUS stack: A4/A9/B4/B9
J1 GND stack: A1/A12/B1/B12; shield S1 grounded

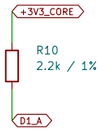
USB FS: embedded driver includes termination; direct D+/D− routes
Pair all four connector D contacts correctly for cable orientation
Ground-only ESD avoids signal-to-VBUS steering backfeed
USB total direct capacitance and protection reviewed on power sheet

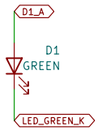


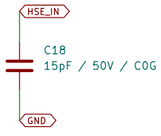



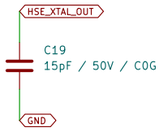


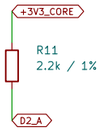


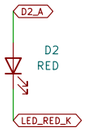



8 MHz fixed HSE for stock F7X2 firmware
15 pF loading assumes approximately 2.5 pF stray
Measure startup, drive and frequency on first article
BOOT pulldown R2 = 2.2k to meet switch minimum wetting current

LED pair combined sink <3 mA
PC14 green, PC15 red
















ENGINEERING DEVELOPMENT — NOT FOR FABRICATION OR FLIGHT

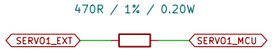
Sheet: /USB + Clock + Service/
File: usb-clock.kicad_sch

**Title: F722 HELICOPTER CONTROLLER / USB + CLOCK + SERVICE**

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-04 | **Rev: A-dev** |
| KiCad E.D.A. 10.0.6 | | Id: 3/9 |



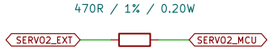

J2
S1 / GND–VX–S







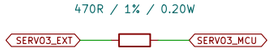
J3
S2 / GND–VX–S

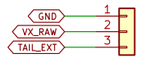




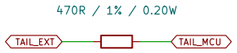


J4
S3 / GND–VX–S




J5
TAIL / GND–VX–S







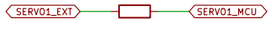
J6
ESC / GND–VX–S







J7
RPM / GND–VX–S

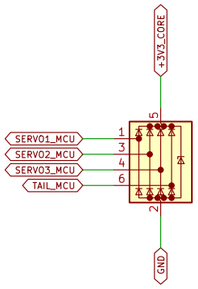




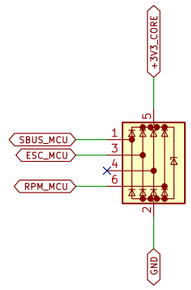


J8
SBUS / GND–VX–S

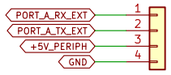



U12
USBLC6-4SC6 / header clamp




U13
USBLC6-4SC6 / header clamp

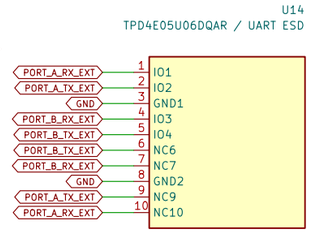



J9
PORT A / JST GH

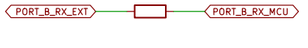


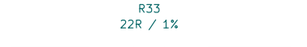


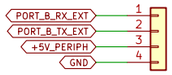




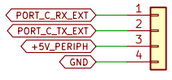

U14 routing lands: NC10/IO1, NC9/IO2, NC7/IO4, NC6/IO5.
External PCB copper is required; active IO pads remain the clamps.




J10
PORT B / JST GH

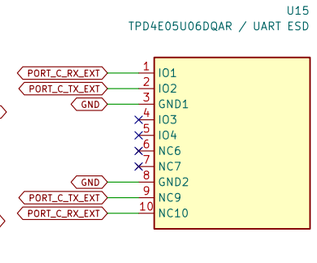


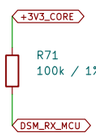

J11
PORT C / JST GH



















J12
DSM / JST ZH

RAW BEC BUS: 5–12.6 V; servo power is NOT regulated 5 V
All header center contacts are common VX_RAW
470R parts rated 0.20 W; RPM/SBUS use HV level translation
Header numbering follows lowest GND / middle VX / upper SIGNAL
Manual THT installation or assembler consignment required

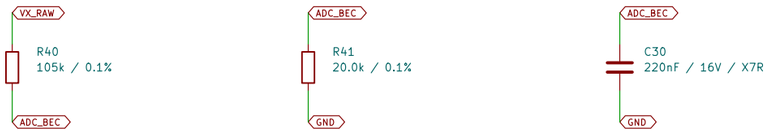
ABC: 5 V / 2 A aggregate; GH <=1 A per contact pair
DSM: nominal 3.3 V / 0.5 A on BEC
USB has a separate lower total current budget
All JST sockets are top–entry as on the reference

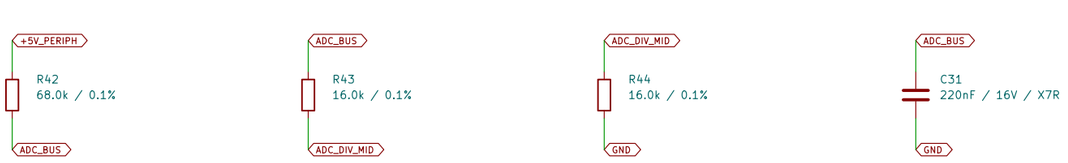
ENGINEERING DEVELOPMENT — NOT FOR FABRICATION OR FLIGHT

Sheet: /io–ports/
File: io–ports.kicad_sch

**Title: F722 HELICOPTER CONTROLLER / EXPOSED PORTS**

| Size: A3 | Date: 2026–10–04 | **Rev: A–dev** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 4/9 |




Stock NEXUS_F7 ADC settings are preserved without custom firmware
BEC on PC1: 20 / (105+20) = 0.160, scale1000/divider160
5 V on PC2: (16+16) / (68+16+16) = 0.320, scale1000/divider320
Ratios use 0.1% resistors; firmware ADC calibration still requires measurement

Measure RAW BEC before electronic protection/mux
Measure peripheral 5 V after source mux
Both RC filters are deliberately slow supply monitors
These inputs are not main propulsion-battery voltage inputs

ENGINEERING DEVELOPMENT – NOT FOR FABRICATION OR FLIGHT

Sheet: /voltage-sense/
File: voltage-sense.kicad_sch

**Title: F722 HELICOPTER CONTROLLER / SUPPLY MONITORING**

| Size: A3 | Date: 2026-10-04 | **Rev: A-dev** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 5/9 |

U5
TPS259472ARPWR

VX_RAW
D9
SMF15CA / bidirectional
C51
1uF / 25V / X7R
R50
1.5M / 1%
EFUSE_EN
EFUSE_EN
IN
EN_UVLO
OVCSEL
PGTH
GND
ILM
DVDT
OUT
PG
ITIMER
EFUSE_OVC
EFUSE_OVC
R51
390k / 1%
EFUSE_ILM
EFUSE_ILM
R52
750R / 1%
EFUSE_DVDT
EFUSE_DVDT
C50
3.3nF / 50V / C0G
C52
1uF / 25V / X7R
VX_PROTECTED
R53
4.7k / 1%
GND

Electronics branch protection only: raw servo bus bypasses U5
13.8 V nominal clamp; about 4.45 A electronic current limit
3.3 nF sets about 0.6 V/ms startup slew
Cold reverse input is not arbitrary hot-reversal immunity

PG selects BEC only once 5 V converter is ready
2 A is total A+B+C allowance on BEC
Verify effective capacitor values, transients and heat
USB supply is current-limited at a much lower total

L1
1.5uH / XGL4020-152MEC
ABC_L1
ABC_L2

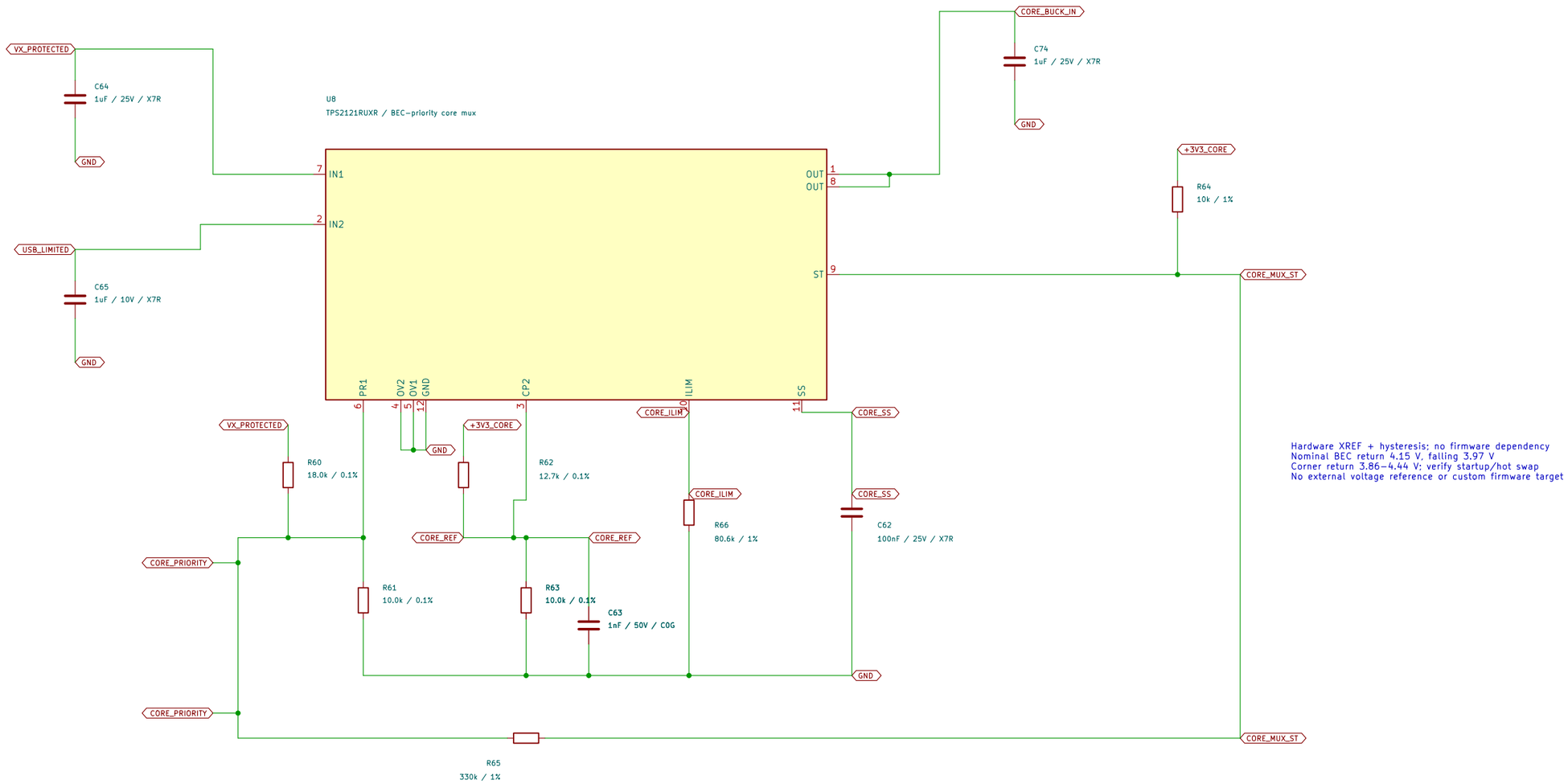
U6
TPS63070RNMR / nominal 5.088V
VIN
VIN
EN
PS_SYNC
GND
FB2
PGND
VSEL
VAUX
L1
L2
VOUT
VOUT
PG
FB
VX_PROTECTED
C53
22uF / 25V / X5R
C54
22uF / 25V / X5R
C71
22uF / 25V / X5R
C72
10uF / 25V / X5R / HF
R57
100k / 1%
ABC_EN
ABC_EN
ABC_VAUX
ABC_VAUX
C58
100nF / 16V / X7R
R56
100k / 1%
ABC_PG
R54
53.6k / 0.1%
ABC_FB
ABC_FB
R55
10.0k / 0.1%
C55
47uF / 10V / X5R
C56
47uF / 10V / X5R
C57
47uF / 10V / X5R
+5V_BEC
C73
10uF / 25V / X5R / HF
R58
47k / 1%

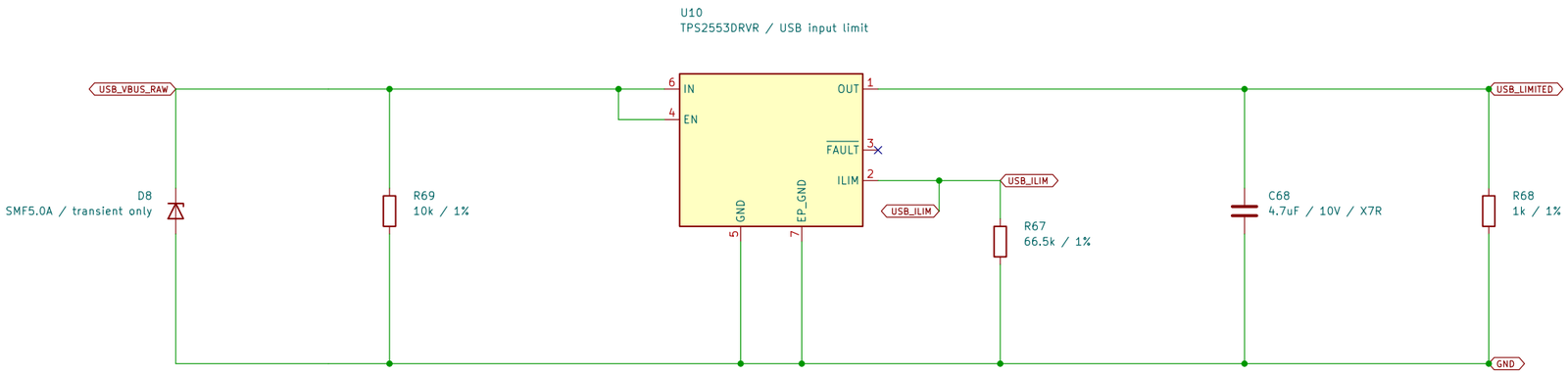
+5V_BEC
C59
1uF / 10V / X7R
U7
TPS21170DRLR / ABC priority mux
VIN1
VIN2
PR1
MODE
GND
ST
VOUT
ABC_PG
USB_LIMITED
C60
2.2uF / 10V / X7R(Murata)
C75
47uF / 10V / X5R
C61
4.7uF / 10V / X7R
+5V_PERIPH

U7 stacked OUT pins: 2 and 7

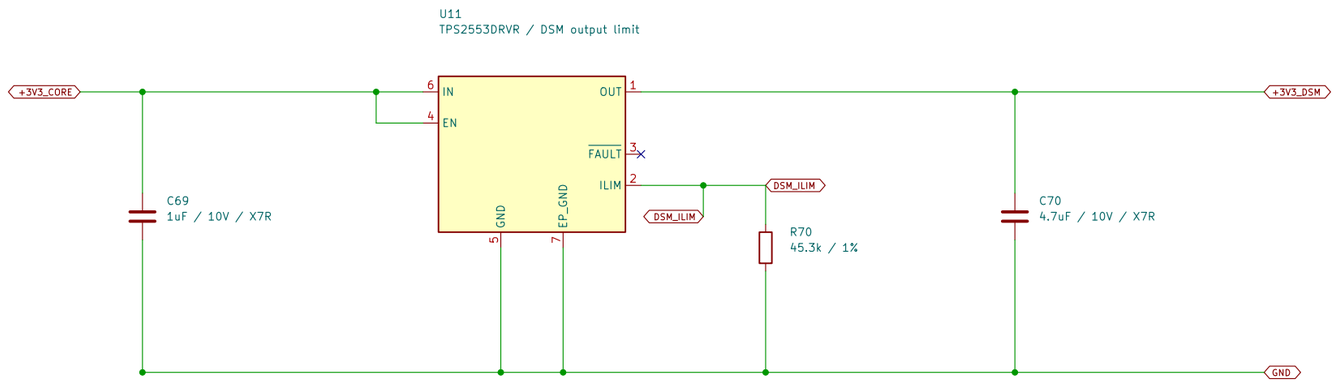
ENGINEERING DEVELOPMENT — NOT FOR FABRICATION OR FLIGHT

| | | |
|---|---|---|
| Sheet: /power-bec/ | | |
| File: power-bec.kicad_sch | | |
| **Title: F722 HELICOPTER CONTROLLER / BEC + PERIPHERAL 5V** | | |
| Size: A3 | Date: 2026-10-04 | **Rev: A-dev** |
| KiCad E.D.A. 10.0.6 | | Id: 6/9 |


VX_PROTECTED
C64
1uF / 25V / X7R
GND
USB_LIMITED
C65
1uF / 10V / X7R
GND
U8
TPS2121RUXR / BEC-priority core mux
IN1
IN2
OUT
OUT
ST
PR1
OV2
OV1
GND
CP2
ILIM
SS
CORE_BUCK_IN
C74
1uF / 25V / X7R
GND
+3V3_CORE
R64
10k / 1%
CORE_MUX_ST
VX_PROTECTED
R60
18.0k / 0.1%
+3V3_CORE
R62
12.7k / 0.1%
GND
CORE_ILIM
CORE_SS
CORE_ILIM
R66
80.6k / 1%
CORE_SS
C62
100nF / 25V / X7R
CORE_REF
CORE_REF
CORE_PRIORITY
R61
10.0k / 0.1%
R63
10.0k / 0.1%
C63
1nF / 50V / C0G
GND
CORE_PRIORITY
CORE_MUX_ST
R65
330k / 1%
Hardware XREF + hysteresis; no firmware dependency
Nominal BEC return 4.15 V, falling 3.97 V
Corner return 3.86–4.44 V; verify startup/hot swap
No external voltage reference or custom firmware target
U9
TPS62162DSGR / 3.3V 1A
L2
3.3uH / DFE322512F
CORE_BUCK_IN
VIN
EN
PG
SW
VOS
FB
PAD
PGND
AGND
CORE_SW
CORE_SW
+3V3_CORE
C66
22uF / 25V / X5R
GND
GND
C67
22uF / 10V / X7R
GND
0.30 A provisional core + 0.50 A DSM budget
EP9 = GND; direct VOS sensing at C67
Thermal via-in-pad only with filled/capped process
Check sensor power-ramp limits at actual pins
Power flags below mark externally supplied rails or passive-filtered regulator outputs
They are ERC source declarations, not missing physical regulators
GND
VX_RAW
USB_VBUS_RAW
+3V3_CORE
+3V3_IMU
+3V3_ANALOG
ENGINEERING DEVELOPMENT — NOT FOR FABRICATION OR FLIGHT
Sheet: /power-core/
File: power-core.kicad_sch
Title: F722 HELICOPTER CONTROLLER / CORE SOURCE MUX + 3V3
Size: A3
Date: 2026-10-04
Rev: A-dev
KiCad E.D.A. 10.0.6
Id: 7/9




C20 on USB sheet is the direct VBUS input bypass; place near U10
Settled USB limit 351 / 397 / 449 mA min/typ/max
Receiver loads share this budget with the controller
Do not advertise USB as the full BEC-mode power rating



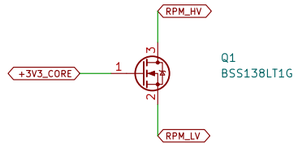

DSM settled limit 519 / 577 / 644 mA min/typ/max
Minimum limit exceeds the 0.5 A specified load
3.3 V is nominal; regulator tolerance and switch drop apply
No shared ABC regulator overload from DSM load

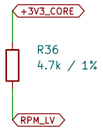
ENGINEERING DEVELOPMENT — NOT FOR FABRICATION OR FLIGHT

Sheet: /power-usb-dsm/
File: power-usb-dsm.kicad_sch

**Title: F722 HELICOPTER CONTROLLER / USB + DSM CURRENT LIMITS**

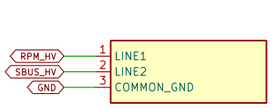
| Size: A3 | Date: 2026-10-04 | **Rev: A-dev** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 8/9 |

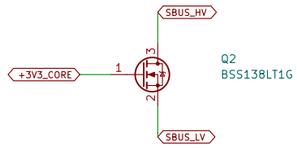


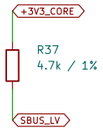













External signal -> 470R on port sheet -> HV drain
Gate = core supply; source pulled to the same core supply
220R then original MCU node, protected by U13
No fixed inverter: frequency/serial polarity is preserved
Stock frequency-input pull-up never sees a VX-level high

Qualify real cable/device edges, startup and powered-off states
No extra source capacitor fitted
No claim of sustained negative-signal or wrong-output-wiring immunity
ESC and tail bidirectional DShot paths remain unbuffered

ENGINEERING DEVELOPMENT – NOT FOR FABRICATION OR FLIGHT

Sheet: /input-levels/
File: input-levels.kicad_sch

**Title: F722 HELICOPTER CONTROLLER / RPM + SBUS INPUT LEVELS**

| Size: A3 | Date: 2026-10-04 | **Rev: A-dev** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 9/9 |